

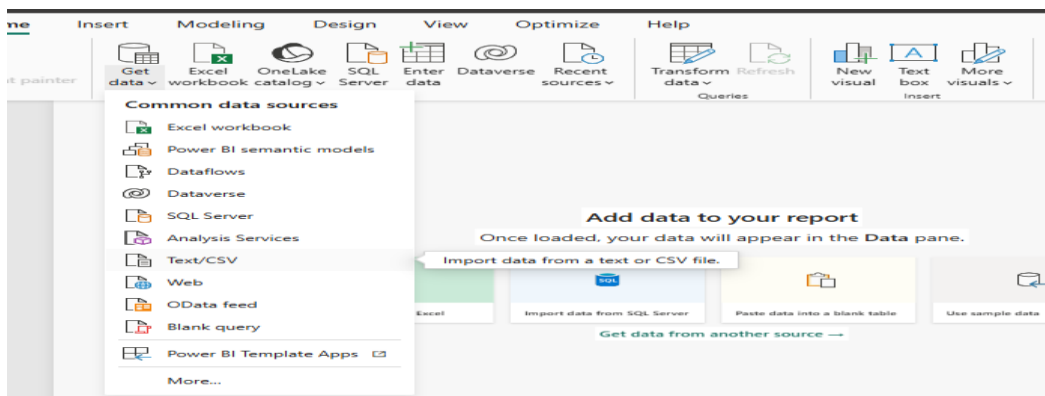
Practical No. 2

Aim: Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sqlserver / Power BI.

The three stages of ETL are:

1. Extraction
Data is collected from one or more data sources such as Excel files, CSV files, databases or APIs.
2. Transformation
The extracted data is cleaned and modified according to analytical requirements. Common transformations include:
 - Removing duplicate records
 - Handling missing values
 - Changing data types
 - Renaming columns
 - Filtering records
 - Creating calculated columns
 - Categorizing data
3. Loading
The transformed data is loaded into the target system or data model where it can be analyzed and visualized.

Step 1: Import the required CSV datasets into Power BI Desktop.



Step 2: Import the Products dataset and preview the product data.

The screenshot shows the 'Import data from a text or CSV file' dialog box in Power BI Desktop. The 'File Origin' is set to 'Text/CSV'. The 'Delimiter' is set to 'Comma'. The 'Data Type Detection' is set to 'Based on first 200 rows'. The 'Table Name' is 'products'. The 'Table Structure' is shown as follows:

ProductID	ProductName	SupplierID	CategoryID	QuantityPerUnit	UnitPrice	UnitsInStock	UnitsOnOrder	ReorderLevel
1	Chai	1	1	10 boxes x 20 bags	18	39	0	0
2	Chang	1	1	24 - 12 oz bottles	19	17	40	0
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10	13	70	0
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22	53	0	0
5	Chef Anton's Gumbo Mix	2	2	36 boxes	22	25	0	0
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25	120	0	0
7	Uncle Bob's Organic Dried Pears	3	2	12 - 1 lb pkgs.	30	15	0	0
8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40	6	0	0
9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97	29	0	0
10	Ikura	4	6	12 - 200 ml jars	31	31	0	0
11	Queso Cabrales	5	4	1 kg pkg.	21	22	30	0
12	Queso Manchego La Pastora	5	4	10 - 500 g pkgs.	38	86	0	0
13	Konbu	6	6	2 kg box	6	24	0	0
14	Tofu	6	7	40 - 100 g pkgs.	23	35	0	0
15	Genen Shoyu	6	2	24 - 250 ml bottles	15	39	0	0
16	Pavlova	7	3	32 - 500 g boxes	17	29	0	0
17	Alice Mutton	7	6	20 - 1 kg tins	39	0	0	0
18	Carnarvon Tigers	7	6	16 kg pkg.	62	42	0	0
19	Teatime Chocolate Biscuits	8	3	10 boxes x 12 pieces	9	25	0	0
20	Sir Rodney's Marmalade	8	3	30 gifts boxes	81	40	0	0

Step 3: Retain the required Products columns: ProductID, ProductName, QuantityPerUnit and UnitsInStock.

The screenshot shows the Power Query Editor with the 'products' table loaded. The table has the following columns: ProductID, ProductName, SupplierID, CategoryID, QuantityPerUnit, UnitPrice, UnitsInStock, UnitsOnOrder, and ReorderLevel. The 'ProductID' column is highlighted. The 'Table.TransformColumnTypes' function is applied to the table, with the following parameters: {"Promoted Headers", {"ProductID", Int64.Type}, {"ProductName", type text}, {"SupplierID", type text}, {"CategoryID", Int64.Type}, {"QuantityPerUnit", type text}, {"UnitPrice", type text}, {"UnitsInStock", Int64.Type}, {"UnitsOnOrder", Int64.Type}, {"ReorderLevel", Int64.Type}}. The 'ProductID' column is set to 'Int64.Type'. The 'ProductName' column is set to 'type text'. The 'SupplierID' column is set to 'type text'. The 'CategoryID' column is set to 'Int64.Type'. The 'QuantityPerUnit' column is set to 'type text'. The 'UnitPrice' column is set to 'type text'. The 'UnitsInStock' column is set to 'Int64.Type'. The 'UnitsOnOrder' column is set to 'Int64.Type'. The 'ReorderLevel' column is set to 'Int64.Type'. The 'Table.TransformColumnTypes' function is applied to the table, with the following parameters: {"Promoted Headers", {"ProductID", Int64.Type}, {"ProductName", type text}, {"SupplierID", type text}, {"CategoryID", Int64.Type}, {"QuantityPerUnit", type text}, {"UnitPrice", type text}, {"UnitsInStock", Int64.Type}, {"UnitsOnOrder", Int64.Type}, {"ReorderLevel", Int64.Type}}. The 'ProductID' column is set to 'Int64.Type'. The 'ProductName' column is set to 'type text'. The 'SupplierID' column is set to 'type text'. The 'CategoryID' column is set to 'Int64.Type'. The 'QuantityPerUnit' column is set to 'type text'. The 'UnitPrice' column is set to 'type text'. The 'UnitsInStock' column is set to 'Int64.Type'. The 'UnitsOnOrder' column is set to 'Int64.Type'. The 'ReorderLevel' column is set to 'Int64.Type'.

Step 4: Change the data type of UnitsInStock to Whole Number.

Power BI Desktop interface showing the 'Change Type' dialog box for the 'UnitsInStock' column. The dialog is set to 'Whole Number'. The background shows a table with columns: ProductID, ProductName, QuantityPerUnit, and UnitsInStock. The 'UnitsInStock' column is highlighted in green.

ProductID	ProductName	QuantityPerUnit	UnitsInStock
10248	VINET	12	39
10249	TOMSP	6	17
10250	HANAR	4	13
10251	VICTE	3	53
10252	SUPRD	5	0
10253	HANAR	5	120
10254	CHOPS	5	15
10255	CHOPS	5	29
10256	CHOPS	5	31

Step 5: Import the Orders dataset into Power BI.

Power BI Desktop interface showing the 'Orders' dataset imported. The 'Query Settings' pane on the right shows the 'Table.TransformColumns' step. The background shows a table with columns: OrderID, ProductID, and UnitsInStock.

OrderID	ProductID	UnitsInStock
1	10248	39
2	10248	17
3	10248	13
4	10249	53
5	10249	0

Step 6: Import the Order_Details dataset into Power BI.

Power BI Desktop interface showing the 'Order_Details' dataset imported. The 'Query Settings' pane on the right shows the 'Table.TransformColumns' step. The background shows a table with columns: OrderID, ProductID, Customer, and UnitsInStock.

OrderID	ProductID	Customer	UnitsInStock
1	10248	VINET	39
2	10249	TOMSP	17
3	10250	HANAR	13
4	10251	VICTE	53
5	10252	SUPRD	0
6	10253	HANAR	120
7	10254	CHOPS	15

Step 7: Merge the Orders and Order_Details queries using the common OrderID field.

Power BI Desktop interface showing the 'Merge' dialog box. The 'Join Kind' is set to 'Left Outer (all from first, matching from second)'. The background shows a table with columns: OrderID, CustomerID, EmployeeID, OrderDate, RequiredDate, ShippedDate, ShipVia, and UnitsInStock.

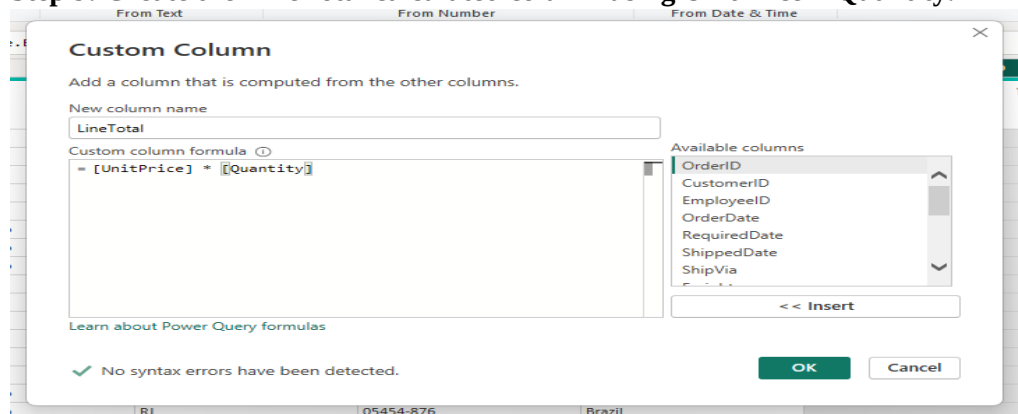
OrderID	CustomerID	EmployeeID	OrderDate	RequiredDate	ShippedDate	ShipVia	UnitsInStock
10248	VINET	5	04-07-1996 00:00:00	01-08-1996 00:00:00	16-07-1996 00:00:00	3	39
10249	TOMSP	6	05-07-1996 00:00:00	16-08-1996 00:00:00	10-07-1996 00:00:00	1	17
10250	HANAR	4	08-07-1996 00:00:00	05-08-1996 00:00:00	12-07-1996 00:00:00	2	13
10251	VICTE	3	08-07-1996 00:00:00	05-08-1996 00:00:00	15-07-1996 00:00:00	1	53

Step 8: Expand Order_Details and select ProductID, UnitPrice and Quantity.

Power BI Desktop interface showing the 'Expand' dialog box. The 'Expand' option is selected. The background shows a table with columns: OrderID, ProductID, UnitPrice, Quantity, and Discount.

OrderID	ProductID	UnitPrice	Quantity	Discount
10248	42	9.8	12	0
10248	72	34.8	5	0
10249	14	18.6	9	0
10249	51	42.4	40	0

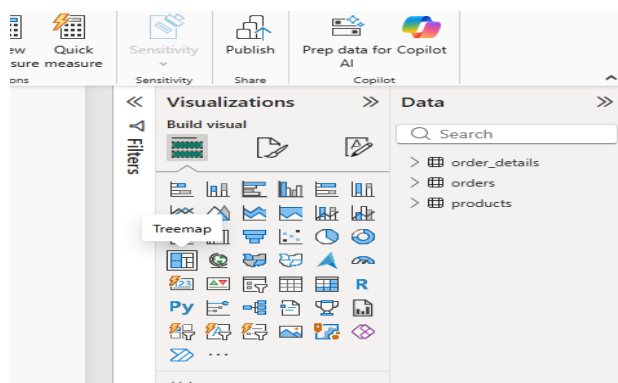
Step 9: Create the LineTotal calculated column using UnitPrice × Quantity.



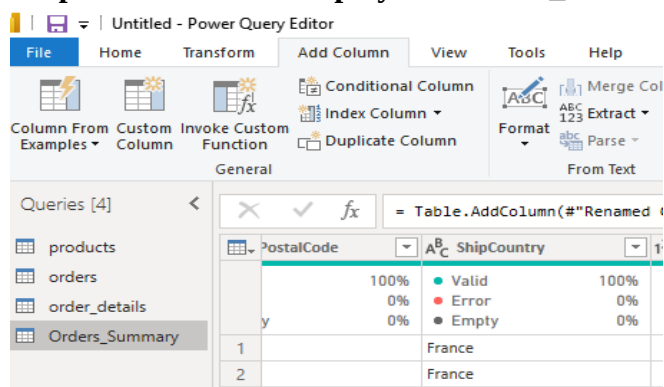
Step 10: Verify the transformed data with the calculated LineTotal column.

ProductID	UnitPrice	Quantity	LineTotal
11	14	12	168
42	9.8	10	98
72	34.8	5	174
14	18.6	9	167.4
51	42.4	40	1696
41	7.7	10	77
51	42.4	35	1484
65	16.8	15	252
22	16.8	6	100.8
57	15.6	15	234
65	16.8	20	336
20	64.8	40	2592

Step 11: Verify that the Orders, Order_Details and Products tables are loaded successfully in Power BI Data view.



Step 12: Create and display the Orders_Summary query in Power Query.



Step 13: Verify Orders_Summary along with the required tables in the Queries/Data model.

Close	New Query	Data Sources	Parameters	Output Data	Query	Man
Queries [4]						
products orders order_details Orders_Summary						
✕ ✓ fx = Table.Distinct(#"Removed Other Columns", {"OrderID"})						
123 OrderID OrderDate						
Valid Error Empty 100% 0% 0%						
Valid Error Empty 100% 0% 0%						
1	10248	04-07-1996 00:00:00				
2	10249	05-07-1996 00:00:00				
3	10250	08-07-1996 00:00:00				
4	10251	08-07-1996 00:00:00				
5	10252	09-07-1996 00:00:00				
6	10253	10-07-1996 00:00:00				
7	10254	11-07-1996 00:00:00				
8	10255	12-07-1996 00:00:00				
9	10256	15-07-1996 00:00:00				
10	10257	16-07-1996 00:00:00				
11	10258	17-07-1996 00:00:00				
12	10259	18-07-1996 00:00:00				
13	10260	19-07-1996 00:00:00				
14	10261	19-07-1996 00:00:00				
15	10262	22-07-1996 00:00:00				
16	10263	23-07-1996 00:00:00				
17	10264	24-07-1996 00:00:00				
18	10265	25-07-1996 00:00:00				

Step 14: View the final Power BI data model and verify the relationships between the tables.

